

Blue for Bob — Virtual Seats Platform: Functional Design

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Target launch: 30 May 2026 ("Blue for Bob Day", Edgbaston)

1. Product summary

A web app that turns a stylised 2D plan of Edgbaston into a fundraising mechanic: supporters pick a seat, leave a tribute, build an avatar, donate (£10 minimum), and get a shareable seat card. Every donation enters a prize draw. The stadium fills up live as donations come in, creating a visible "we did this together" moment for Blue for Bob Day.

Inspiration: McGrath Foundation Pink Test. Same emotional mechanic, our charity, our event, our seat map.

2. Locked-in decisions (post-review with Mark)

Decision	Choice	Rationale
Pricing	£10 floor + optional bump (£25 / £50 / custom)	Honours brief intent; lifts AOV without forcing tier choice
Identity model	Enthuse pattern: identify yourself OR donate anonymously, both valid	Industry-standard for UK charity giving. Two independent privacy toggles: hide amount, hide name
Avatar builder	Full builder in v1	Approved by Mark — emotional payoff worth the build cost
Prize draw	Full draw engine with audit log	Approved by Mark — needed for governance & repeatability
Hosting	Build for both standalone and iframe	App must run at <code>seats.bobwillisfund.org</code> and inside the existing WordPress iframe pattern
Gift Aid	In v1 — full HMRC declaration step	Charity cannot afford to skip 25% uplift
Repo	<code>markmusson/bwf-virtual-seats</code> (transferable)	Fastest path; transfer to BWF org if/when they want it
Timeline	Full v1 by 30 May	Tight. Coding starts week of 4 May. No buffer for unknowns

3. The stadium

The mockup at `bobwillisfund.org/virtualseats` defines our model. It is a 2D top-down rendering of Edgbaston, six stands, drawn as concentric arcs on an HTML5 canvas. Stands and seat counts (from the source):

Stand	Sub	Tier (mock)	Rows
Eric Hollies Stand	The Famous Stand	Standard	13
Wyatt Stand	RES Wyatt	General	7
Stanley Barnes	East Side	Standard	8
South Stand	Pavilion End	Premium	10
West Stand	—	Standard	11
Priory Stand	Family Stand	General	5

Total seats land in the low thousands once seats-per-row is computed by arc length. We keep this geometry — it works, it loads fast, and Adam's team has signed off on the look.

Brief vs. mock divergence: The brief says "3D digital version of Edgbaston". The mock is 2D top-down. **Recommendation: keep the 2D plan.** It's faster to render, mobile-friendly, accessible, and it's what Adam's team has already iterated on. A 3D fly-through stadium is a v2 wow-factor, not a launch requirement. I'll flag this back to Adam to confirm.

3.1 Seat states

- **Empty** — light blue, claimable
- **Claimed** — dark blue or filled with avatar
- **Held** — yellow (someone else has it in checkout)
- **Selected** — gold pulse (this user)
- **Disabled** — grey (admin-locked, e.g., reserved for a campaign sponsor)

3.2 Live filling

Supabase Realtime broadcasts each new claim. Other browsers see the stadium fill up live without refresh. This is the single biggest "feel" moment of the product — when donors see the room around their seat fill while they're paying, the whole thing comes alive.

4. User journeys

4.1 Primary journey: anonymous supporter donates

1. Land on  — hero image, Bob's story snippet, live progress bar (£X raised, N seats filled, M to go), big "Find your seat" CTA.
2. Stadium loads. Pan/zoom on mobile. Tap any empty seat.
3. Modal opens with **6-step wizard** (Enthuse pattern):
 1. **Amount** — £10 default tile, plus £25 / £50 / custom; two privacy toggles ("Hide my donation amount" / "Donate anonymously").
 2. **Details** — name, email; if anonymous: just email for receipt.
 3. **Tribute** — display name (defaults to first name), optional dedication message (max 140 chars), in-memory toggle.
 4. **Avatar** — build it: skin, hair, cap colour, shirt (cricket whites / blue jersey / hoodie), accessory. Or skip and use a default Blue dot.
 5. **Gift Aid** — UK taxpayer? Yes/no. If yes: full HMRC declaration text + address fields + tickbox.
 6. **Pay** — Stripe Embedded Checkout. Apple Pay / Google Pay / card. £10 + bumps + Gift Aid info shown.

4. Stripe processes. Webhook fires. Seat is claimed. User redirected to `/seat/[id]`.
5. Confirmation page: their avatar in their seat, share buttons (X, WhatsApp, Facebook, copy link), prize draw entry confirmation, "donate again" / "find another seat" CTA.

4.2 Returning donor (magic link)

The receipt email contains a magic link to `/manage`. Clicking it logs them in (Supabase magic-link auth). They can:

- See all their claimed seats
- Edit display name / tribute (within 24 hours of donation, or always for non-anonymous donors)
- Re-download their seat card
- Re-share

4.3 Tribute wall

`/wall` — a chronological feed of all non-anonymous tributes. Each card shows seat, avatar, name, message. Helps SEO, helps emotional pull, drives more conversions. Anonymous tributes show as "A friend" with no avatar identity.

4.4 Admin (BWF staff)

`/admin` — basic auth, separate role. Adam, Ricky, Mark.

- Live counters (raised, claimed, gift-aid-eligible, prize entries)
- Per-stand breakdown
- Recent transactions table
- Tribute moderation queue (auto-flagged by profanity filter; manual approve/reject)
- Disable a seat (e.g., refund triggered)
- Run prize draw
- Export donor CSV (Stripe customer ID, email, name, address, gift aid Y/N, amount, seat) for thank-you flows
- Pricing controls (premium/standard/general overrides — already prototyped in mock)

5. Avatar system

Goal: every claimed seat becomes a tiny human in cricket-day kit.

Tech approach: SVG sprite atlas, configuration stored as JSON, rendered client-side. Do **not** store rendered PNGs — too much storage, too slow.

```
{
  "skin": 2,
  "hair": "buzz-cut",
  "hairColor": "#3a2a18",
  "top": "cricket-whites",
  "topColor": "#ffffff",
  "cap": "navy",
  "accessory": "sunglasses"
}
```

Builder UI: wheel-pickers / horizontal scroll on mobile, side-by-side panels on desktop. Live preview to the right. Should feel like Apple's Memoji-lite — fast, fun, ~30 seconds of joy.

Render cost: at 12px on the seat map, you can't see fine detail. We render simplified avatars on the canvas (just skin + top + cap), then the full sprite on the seat card and confirmation screen.

Library options:

- DiceBear (open source, multiple styles, can self-host) — recommended starting point
- Custom SVG kit by BWF designer — better brand fit but adds a week+

Recommendation: start with DiceBear "lorelei" or "pixel-art" tweaked to BWF colours, swap to custom kit in v1.1 if budget allows.

6. Prize draw engine

UK prize-draw rules require either skill-based entry, free entry route, or full lottery licence. Charities sidestep this if there's a free entry route. **Hard requirement: free entry route by post must exist.** Adam to provide the postal address for the T&Cs page.

6.1 Mechanic

- Every paid donation = 1 entry. Bumps up donations don't grant extra entries (keep it simple, defensible, fair).
- Free entries (postal) are added manually via the admin panel.
- One prize, one winner. Toyota car or whatever Adam confirms.

6.2 Draw process

1. Campaign closes. Admin clicks "Run draw".
2. Server-side: pull all `prize_entries` where `eligible = true`. Use a CSPRNG (Node `crypto.randomInt`) to pick one. Persist `prize_draw_results` row with seed, timestamp, winning entry, total entry count.
3. Email the winner. Display public proof (entry count, draw timestamp, winning seat ID — not PII) on the T&Cs page.

6.3 T&Cs page

`/prize-terms` — Adam writes copy, we host. Includes:

- Prize details
- Eligibility (UK residents 18+, no employees)
- Free entry route
- Closing date / draw date
- How winner is contacted
- Promoter (BWF charity number)

7. Share cards

Every claimed seat gets a unique URL: `/seat/[id]`.

7.1 OG image

Generated dynamically by `@vercel/og` (React-based PNG generator). Includes:

- BWF logo
- Donor's avatar (or anonymous placeholder)
- Seat tag ("Eric Hollies Stand · Row 3, Seat 17")
- Tribute message if non-private
- "I'm at Blue for Bob Day. Are you?" overlay

- Edgbaston silhouette

7.2 Share buttons

Native Web Share API on mobile (one-tap to anywhere installed). Fallback buttons for desktop: X, WhatsApp, Facebook, Copy link.

7.3 Downloadable seat card

PNG export of the same image, slightly higher res, suitable for printing or saving to camera roll.

8. Architecture (high level)

```
[ Browser ]
|
| (HTTPS, optional iframe parent)
v
[ Vercel Edge / Next.js App Router ]
|  pages/* (server components, SSR for /, /wall)
|  api/* (route handlers — checkout, webhook, admin actions)
|  /seat/[id]/og.png (Vercel OG dynamic image)
|
+-- [ Stripe Checkout ] (hosted/embedded)
|    webhook --> /api/stripe/webhook
|
+-- [ Supabase ]
|    Postgres (seats, claims, donors, holds, entries, audit)
|    Auth (magic link for /manage and /admin)
|    Realtime (broadcast new claims)
|    Storage (avatar sprite atlas, BWF assets)
|
+-- [ Resend / Postmark ] (transactional email)
```

Stack:

- **Next.js 15** (App Router, Server Components, Edge runtime where possible)
- **TypeScript** strict
- **Tailwind v4** + a tiny set of primitives — no UI library bloat
- **Supabase** Postgres + Auth + Realtime
- **Stripe** Embedded Checkout + Webhooks
- **Resend** for transactional email
- **Plausible** (or Vercel Analytics) for funnel
- **Sentry** for error tracking

9. Data model (logical)

Table	Purpose
<code>seats</code>	One row per seat: id, stand, row, col, x_coord, y_coord, tier, base_price_pence, status
<code>donors</code>	One per email: id, email, full_name, address fields (for Gift Aid), is_uk_taxpayer, marketing_opt_in, created_at
<code>claims</code>	The canonical "this seat belongs to this donor": seat_id, donor_id, donation_id, display_name, tribute_message, is_anonymous, is_amount_private, avatar_config (jsonb), claimed_at
<code>donations</code>	Stripe payment record: id, donor_id, amount_pence, gift_aid, gift_aid_amount_pence, stripe_payment_intent_id, status, created_at
<code>seat_holds</code>	Soft-lock during checkout: seat_id, session_id, expires_at (10 min)
<code>prize_entries</code>	One per paid donation + one per postal entry: id, donation_id (nullable for postal), donor_id, source, eligible, created_at
<code>prize_draw_results</code>	id, drawn_at, winning_entry_id, seed, total_entries
<code>tribute_moderation</code>	claim_id, status (pending/approved/rejected), flagged_by, reviewed_by, reviewed_at
<code>audit_log</code>	Generic admin-action log

Full SQL is in `03-claude-code-prd.md`.

10. Concurrency: the seat-locking problem

Two donors hit the same seat at the same time. We must not double-charge or double-claim.

Pattern:

1. User clicks seat → `POST /api/holds` creates a `seat_holds` row with `expires_at = now() + 10 minutes`. Postgres unique constraint on `(seat_id) WHERE expires_at > now() AND released = false` enforces single-holder.
2. If the constraint fails (someone else has it), the API returns 409 and the UI says "just gone — try another".
3. User completes Stripe Checkout. Webhook (`checkout.session.completed`) consumes the hold, creates the `claim` and `donation`, releases the hold.
4. If user abandons, the hold expires automatically. A nightly cron sweeps expired holds (cosmetic — the constraint already ignores them).

Idempotency: Stripe webhook events are idempotent by `event.id`. We persist seen event IDs and refuse duplicates.

11. Mobile-first design

90%+ of donors will be on mobile (Pink Test data backs this). Every screen must work in portrait, one-handed, on a £100 Android.

- Stadium canvas: pinch to zoom, pan with finger; double-tap to zoom into a stand.
- Wizard: full-bleed cards, big tap targets, system keyboard for inputs, no modals-in-modals.
- Avatar builder: swipe between options, no drag-and-drop.
- Stripe Embedded Checkout natively handles Apple Pay / Google Pay — critical for conversion.

12. Accessibility

- Stadium has a **list view fallback** (`/seats?view=list`) showing stand → row → seat selectors with screen-reader

labels. Same data, no canvas.

- All interactive elements meet WCAG 2.1 AA contrast (BWF blue + white passes).
- Avatar builder has labelled options, not just colour swatches.
- Tribute messages support unicode (emoji, accented characters).

13. Open questions for Adam

1. **3D vs 2D** — confirm we keep the 2D top-down plan from the mock. (Recommendation: yes.)
2. **Prize details** — what's the actual prize? Toyota? When confirmed?
3. **Total donation target** — what's the £ goal we display in the progress bar?
4. **Promotional code support** — do we need any (e.g., partner clubs)?
5. **Tribute moderation** — auto-publish with retroactive moderation, or hold for review? (Recommendation: auto-publish with profanity filter, manual review queue for flagged.)
6. **Postal address for free prize entry route** — required for legal compliance.
7. **BWF charity number** — needed for footer + T&Cs.
8. **Final domain decision** — `seats.bobwillisfund.org` or iframe at `/virtalseats?`
9. **Email sender** — `seats@bobwillisfund.org` or different? Need DKIM/SPF set up.
10. **Existing Stripe account** — Mark already invited per Adam's email. Confirm test/live mode access and whether we use a dedicated product/price or just one-shot Payment Intents (recommendation: Payment Intents — simpler).
11. **Sponsor seats** — does BWF want to give a block of seats to a sponsor (e.g., a brand pays £5k for 50 seats with their logo)?

14. What this design deliberately leaves out

- 3D rendering of the stadium
- Native mobile apps (web-only)
- Recurring donations (the brief is event-based)
- Donor-to-donor messaging
- Live video or stream integration
- NFTs, blockchain, or any other 2022 nonsense

These could be v2. They will not slip into v1.

See `02-brief-evaluation.md` for what's missing from the original brief, and `03-claude-code-prd.md` for the implementation spec ready to feed to Claude Code.